Modern physics

Monday, November 10, 2025 8:55 AM

$c = \lambda f$, $E = hf$ $\Rightarrow E = \frac{hc}{\lambda}$

Blackbody radiation :

- absorb all incoming radiation, re-emit with spectrum based on temp. only
- hotter objects emit more energy, peak at shorter wavelengths

$$\boxed{\lambda_{max} = \frac{2.898 \times 10^{-3}\ m \cdot K}{T}}$$

Rayleigh - Jeans : assumes any continuous amount of energy based on temp.
holds for long wavelengths, diverges for short

$$\boxed{M_\lambda(\lambda, T) = \frac{2\pi c k_B T}{\lambda^4}}$$ Rayleigh Jeans

Planck : Discrete energy levels
Energy emitted / absorbed in packets (quanta) of size hf (one photon)

Planck constant : $h = 6.626 \times 10^{-34}\ J \cdot s = 4.136 \times 10^{-15}\ eV \cdot s$
(since 1eV = 1.602×10^{-19} J)

$$\boxed{M_\lambda(\lambda, T) = \frac{2\pi hc^2}{\lambda^5} = \frac{1}{e^{hc/(\lambda k_B T)} - 1}}$$ Planck's Law

Photoelectric effect :

- electrons ejected from metal surface when light of high frequency strikes
- light hits metal cathode → eject electrons to anode → measure current

Apply stopping potential V_s to find electrons' max KE :

$$\boxed{\underset{V}{eV_s} = \underset{eV}{K_{max}}}$$

Conclusion : Intensity only affects # of emitted electrons, not the energy
No electrons ejected below threshold f_0.

Light consists of photons, each with energy $\boxed{E = hf}$, momentum $\boxed{p = \frac{h}{\lambda}}$

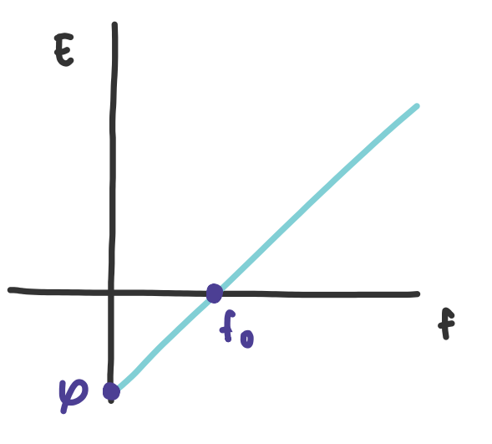
single photon transfers energy to single electron : $\boxed{hf = \underset{\substack{\uparrow \\ \text{work function}}}{\varphi} + K_{max}}$

Photocurrent : $\boxed{I = \underset{C}{(e)} \times \underset{\text{photons/s}}{(rate)} \quad (A)}$

K_{max} increases linearly with f

$$\boxed{f_0 = \frac{\varphi}{h}}$$

Higher intensity = more photons, not more energy per photon



$K = hf - \underset{\substack{\uparrow \\ \text{min energy required} \\ \text{"escape energy"}}}{\varphi}$

X-rays

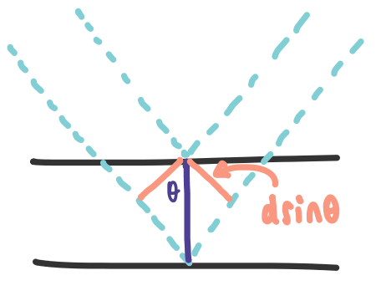
shorter wavelengths than visible light, high photon energy

Bremsstrahlung : Electrons slow down when deflected by nuclei → cont. spectrum energy can be emitted as x-ray
Characteristic x-ray : Incoming electron knocks orbital electron on inner shell → outer shell fills hole + emits discrete x-rays

$$\boxed{\underset{\substack{\uparrow \\ \text{x-ray}}}{E_\gamma} = E_{outer} - E_{inner}}$$

Bragg diffraction

Constructive interference : $\boxed{2d\sin\theta = n\lambda}$ Bragg's Law



Each set of atomic planes reflect x-rays 'in-phase' when Bragg's law satisfied.

Compton Effect

High energy photon (e.g. xray) hits electron → bounce off and lose some energy
like billiard ball

Scattered photon has lower energy, longer wavelength (unexpected by classical physics, where light is a wave so equal frequency)

Explanation: $E = hf$, $p = \frac{h}{\lambda}$
When photon collides with electron, it transfers part of energy and momentum to electron
↳ photon scatters with reduced energy
↳ electron recoils, gaining kinetic energy

$$\boxed{\begin{array}{l} \Delta\lambda = \lambda - \underset{\substack{\uparrow \\ \text{initial wavelength}}}{\lambda_0} = \frac{h}{mc}(1 - \cos\underset{\substack{\uparrow \\ \text{scattering angle} \\ \text{(max at 180°)}}}{\theta}) \\ \frac{h}{mc} = 0.00243\ nm \quad \text{(Compton wavelength)} \end{array}}$$ Compton

Atomic spectra

Emission spectrum : bright coloured lines on dark background
Absorption spectrum : dark lines on continuous spectrum
matching → atoms emit / absorb light at same quantized energy differences

Balmer - Rydberg :
In hydrogen :

$$\boxed{\frac{1}{\lambda} = \underset{\substack{\uparrow \\ R = 1.097 \times 10^{7}\ m^{-1}}}{R}\left(\frac{1}{n'^2} - \frac{1}{n^2}\right) \quad (n > n')}$$ Rydberg

$$\boxed{E = hcR\left(\frac{1}{n'^2} - \frac{1}{n^2}\right)}$$ using $E = \frac{hc}{\lambda}$

Rutherford : Electrons orbit nucleus like planets
Accelerating electrons radiate energy → lose speed → spiral into nucleus → wrong!

Bohr : Electrons move in stationary orbits (discrete), no energy radiated
Energy is quantized (E_1, E_2, ...)
Photon emission / absorption when energy jumps between energy levels ($E_\gamma = E_n - E_{n'}$)

Spectral lines : each transition produces light at specific wavelength $\boxed{\lambda = \frac{hc}{E_\gamma}}$

higher → lower : emits photon w/ energy diff.
lower → higher : absorb photon w/ exact energy diff.

Quantization of angular momentum
$\boxed{L = mvr = n\hbar}$ $(n = 1, 2, 3, \ldots)$ $\boxed{\hbar = \frac{h}{2\pi}}$

centripetal = Coulomb : $\boxed{mv^2 = \frac{ke^2}{r}}$ (force)

$k = 8.99 \times 10^{9}\ N \cdot m^2/C^2$

$E = -\frac{ke^2}{2r}$, $r_n = n^2 a_B$ $\Rightarrow$ $E_n = -\frac{ke^2}{2a_B n^2}$

$$\boxed{E = K + U = \frac{1}{2}mv^2 - \frac{ke^2}{r} = -\frac{ke^2}{2r}}$$ (energy)

Rydberg energy : $E_R = \frac{ke^2}{2a_B} = 13.6eV$

Then, allowed energies are $\boxed{E_n = -\frac{E_R}{n^2} = -\frac{13.6eV}{n^2} = -\frac{m_e k^2 e^4}{2\hbar^2 n^2}}$ (hydrogen atom)

hydrogen radius of n^{th} orbit : $\boxed{r_n = n^2 a_B = \frac{n^2\hbar^2}{m_e k e^2} \quad a_B = 0.0529\ nm}$

Hydrogen-like atoms

Every ke^2 replaced with Zke^2, where Z is atomic number.
(electron orbits nucleus of charge Ze)

$$\boxed{\begin{array}{l} r_n = \frac{n^2 a_B}{Z} = \frac{n^2\hbar^2}{m_e k Z e^2} \\ E_n = -\frac{(13.6eV)Z^2}{n^2} = -\frac{m_e k^2 Z^2 e^4}{2\hbar^2 n^2} \end{array}}$$ (hydrogen-like atom)

Z increase → energy levels more tightly bound (larger -ve), r_n decreases

Louis de Broglie

matter has wave-like properties
$E = hf$, $p = \frac{h}{\lambda}$

a moving particle of mass m and velocity v :

$$\boxed{\underset{\substack{\uparrow \\ \text{de Broglie} \\ \text{wavelength}}}{\lambda} = \frac{h}{p} = \frac{h}{mv} = \frac{h}{\underbrace{\sqrt{2m_e K}}_{p}}}$$

for stable wave, electron wave must fit exactly around orbit's circumference
$\boxed{2\pi r = n\lambda, \quad n = 1, 2, 3, \ldots}$

angular momentum : $L = rp$
$= \left(\frac{n\lambda}{2\pi}\right)\left(\frac{h}{\lambda}\right)$
$= \frac{nh}{2\pi}$
$= n\hbar$

Bohr's model is just a special case of de Broglie's rule